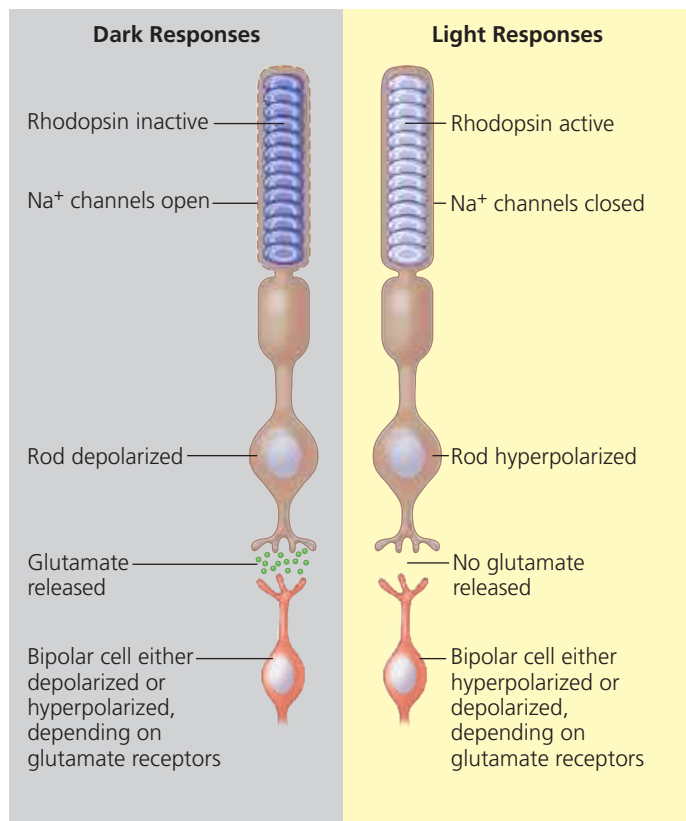


▼ **Figure 50.19 Synaptic activity of rod cells in light and dark.**



? Like rods, cone cells are depolarized when their opsin molecules are inactive. In the case of a cone, why might it be misleading to call this a dark response?

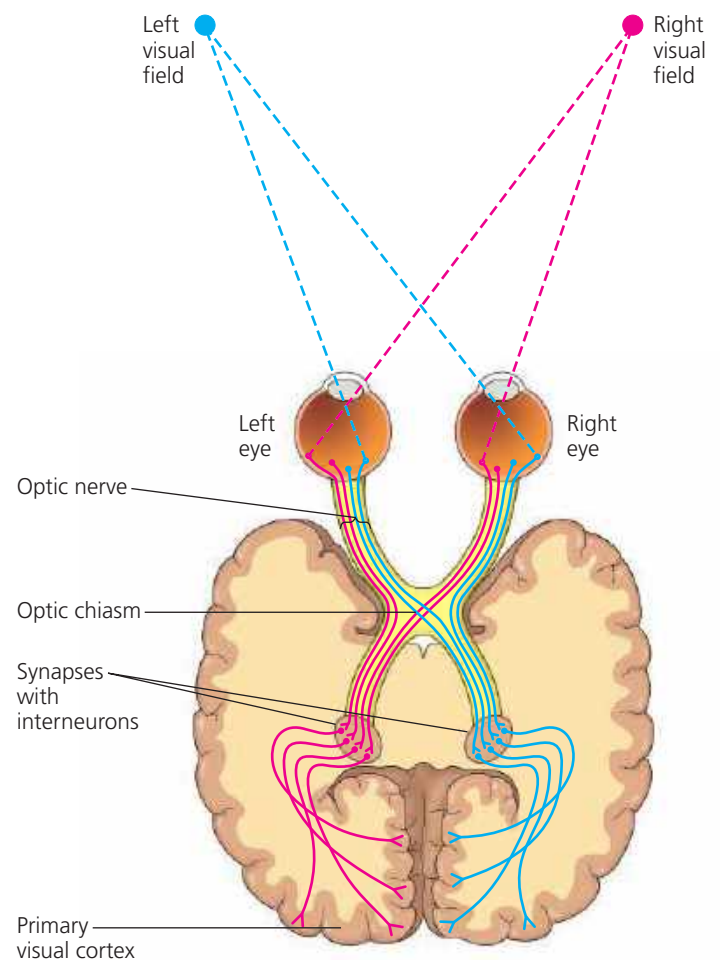
A single ganglion cell receives information from an array of rods and cones, each of which responds to light coming from a particular location. Together, the rods and cones that are feeding information to one ganglion cell define a *receptive field*—the part of the visual field to which that ganglion cell can respond. The fewer rods or cones that supply a single ganglion cell, the smaller the receptive field. A smaller receptive field typically results in a sharper image because the information about where light has struck the retina is more precise.

Processing of Visual Information in the Brain

Axons of ganglion cells form the optic nerves that transmit action potentials from the eyes to the brain (**Figure 50.20**). The two optic nerves meet at the *optic chiasm* near the center of the base of the cerebral cortex. Axons in the optic nerves are routed at the optic chiasm such that sensations from the left visual field are transmitted to the right side of the brain, and sensations from the right visual field are transmitted to the left side of the brain. (Note that each visual field, whether right or left, involves input from both eyes.)

Within the brain, most ganglion cell axons lead to the *lateral geniculate nuclei*, which have axons that reach the *primary visual cortex* in the cerebrum. Additional neurons carry the information to higher-order visual processing and

▼ **Figure 50.20 Neural pathways for vision.** Each optic nerve contains about a million axons that synapse with interneurons in the brain. The nuclei relay sensations to the primary visual cortex, one of many brain centers that cooperate in constructing visual perceptions.



integrating centers elsewhere in the cortex. Researchers estimate that at least 30% of the cerebral cortex, comprising hundreds of millions of neurons in perhaps dozens of integrating centers, takes part in formulating what we actually “see.” Determining how these centers integrate such components of our vision as color, motion, depth, shape, and detail is the focus of much exciting research.

Color Vision

Among vertebrates, most fishes, amphibians, and reptiles, including birds, have very good color vision. Humans and other primates also see color well, but are among the minority of mammals with this ability. For cats and other mammals that are most active at night, a high proportion of rods in the retina is an adaptation that provides keen night vision. Color vision among these nocturnal animals is limited, and they probably see a pastel world during the day.

In humans, the perception of color is based on three types of cones, each with a different visual pigment—red, green, or blue. The three visual pigments, called *photopsins*, are formed